



MidiMovie User Manual

Play a MIDI keyboard along to a video, layer your takes, and export the audio.

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<https://aratus.github.io/MidiMovie/>

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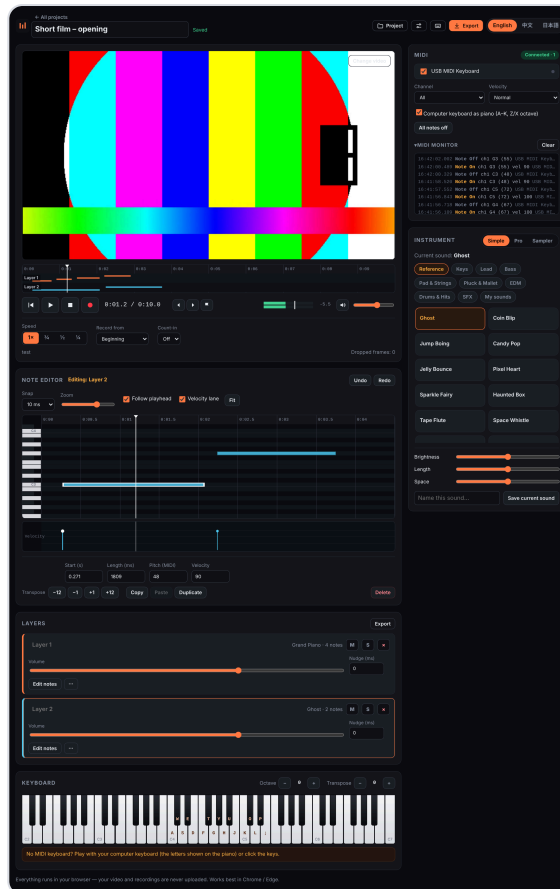
1. What MidiMovie does

MidiMovie runs entirely in your web browser. You load a video, play a MIDI keyboard (or your computer keyboard) while watching it, and the performance is recorded in sync with the picture. Keep the takes you like, stack as many layers as you want, fine-tune every note in a piano-roll editor, and export the result as a WAV audio file.

- Nothing is uploaded: video, notes and sounds stay in your browser.
- Works best in Chrome or Edge (Web MIDI is required for a MIDI keyboard).
- No MIDI keyboard? Play with your computer keyboard or click the on-screen piano.

2. Quick start

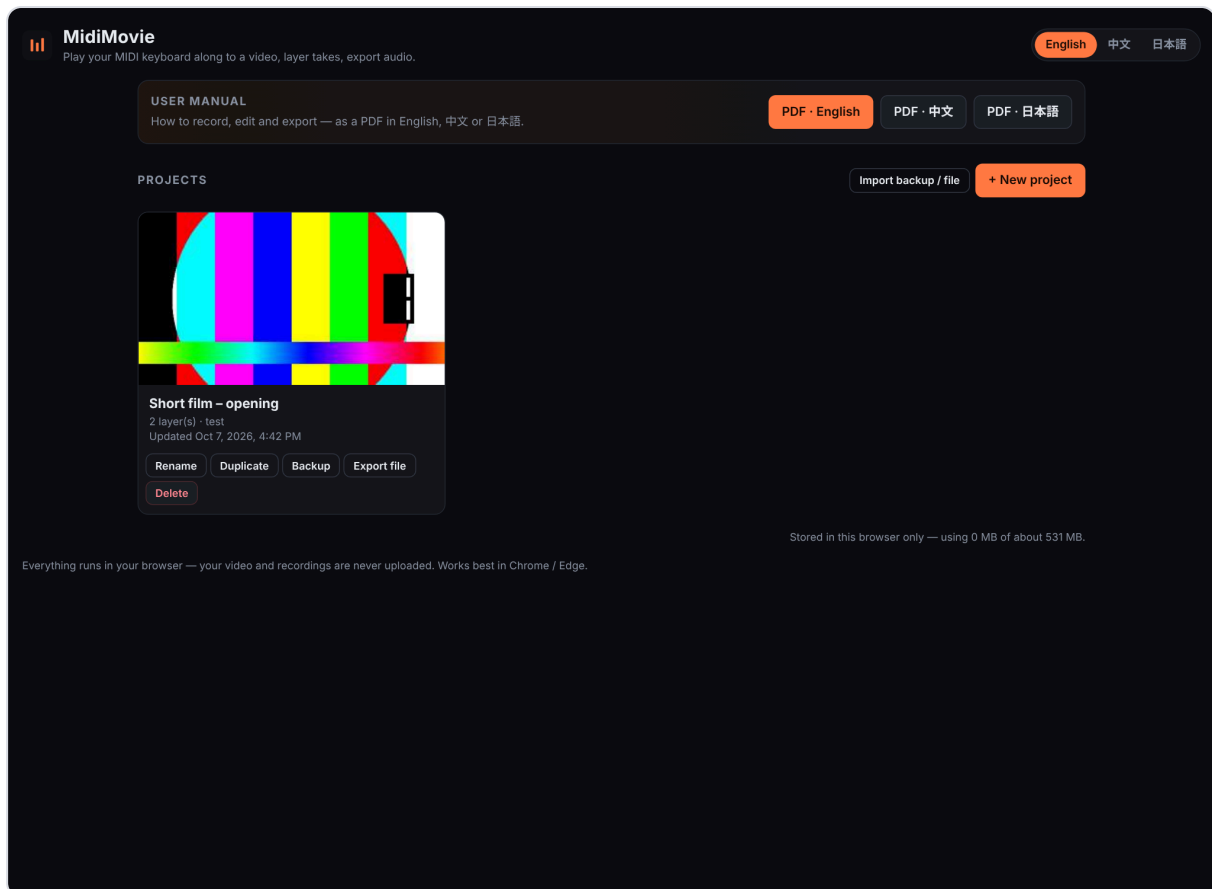
1. Open <https://aratus.github.io/MidiMovie/> and press **+ New project**.
2. Drop a video file onto the player (or press **Load video**).
3. Press **Connect MIDI** and allow access when the browser asks. Skip this step to play with the computer keyboard.
4. Pick a sound in the **Instrument** panel (Simple tab → press a card to hear it).
5. Press the red **Record** button. After the count-in the video plays and your playing is recorded.
6. Press Record again (or Esc) to stop. Choose **Keep as new layer** or **Discard**.
7. Repeat for more layers. Fine-tune notes in the **Note editor**.
8. Press **Export mix (WAV)** to download the audio.



The whole editor screen.

3. Projects and automatic saving

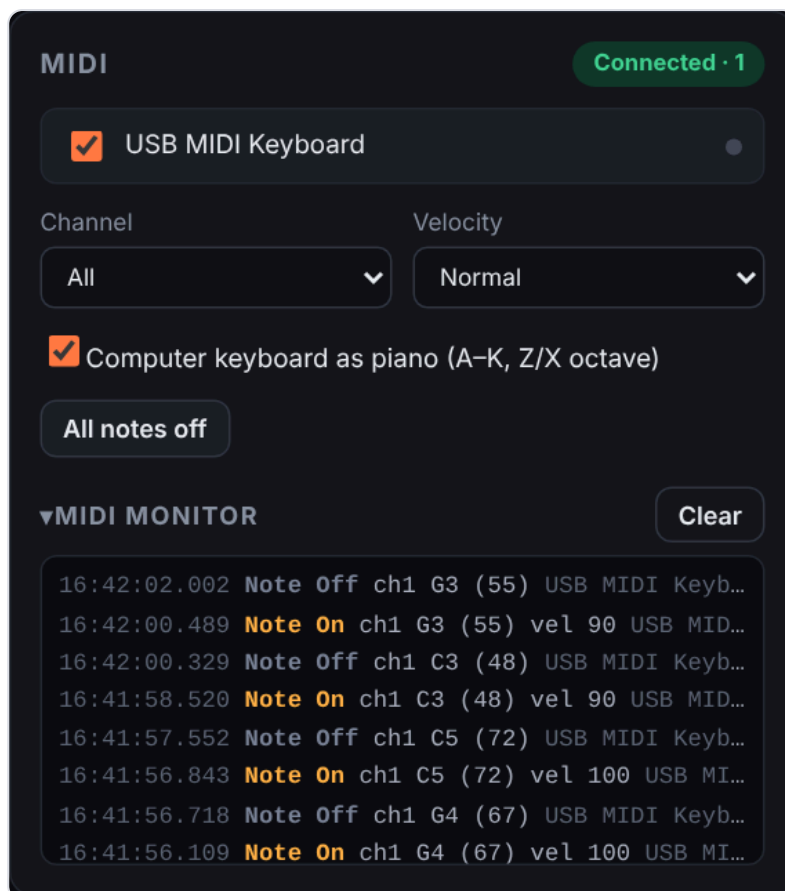
The start page lists your projects. Each project opens the editor with its own video, layers, sounds and settings. Everything is saved automatically a moment after each change; the header shows **Saved** when it is safe.



The project list with the manual links.

- **Reload-safe:** after a reload the layers, sounds, settings, playhead position and the video itself are restored. The video is stored inside your browser (IndexedDB) — no server is involved.
- **Rename** a project from the list or by editing the name at the top of the editor. **Duplicate** copies layers and video. **Delete** removes both.
- **Export file / Import project file** create a backup (.midimovie.json) or move work between computers. The file contains notes and sounds but *not* the video; load the video again after importing.
- Projects live only in the browser and device you used. Clearing site data, private/incognito windows, or a different browser will not show them — export a file for safekeeping.
- Large videos use browser storage; the list page shows how much is used. If the browser refuses to store a video you will see a message and must reload the video after a reload.
- **Backup (ZIP)** in the editor's **Project** dialog (or on each project card) saves the project *with* its video and samples.

4. Connecting a MIDI keyboard



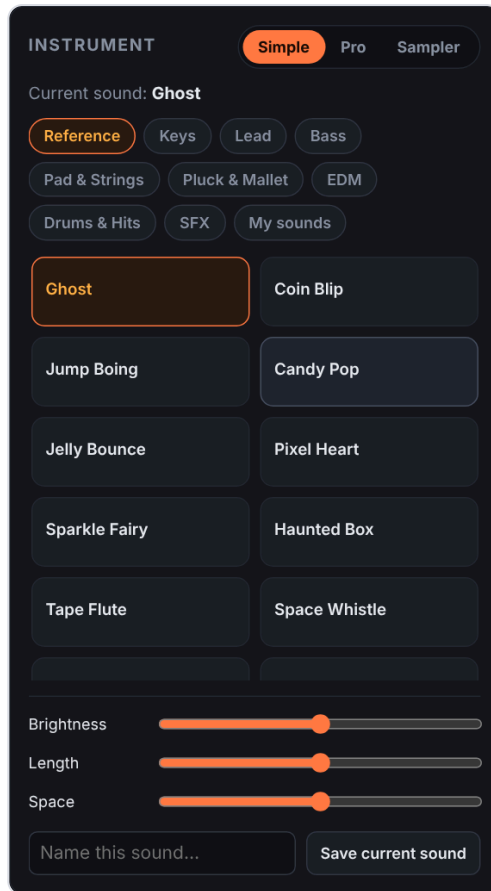
The MIDI panel with the live monitor.

- Plug in your keyboard, press **Connect MIDI** and allow access. Connected inputs appear in a list; untick one to ignore it. Newly plugged devices appear automatically.
- **Channel** limits input to one MIDI channel. **Velocity** → **Fixed (100)** ignores how hard you press.
- The **MIDI monitor** shows incoming messages (Note On/Off, Control Change, Pitch Bend ...) — handy to check that your device works.
- Supported: notes with velocity, sustain pedal (CC 64), pitch bend (recorded too), All Notes Off.
- **Octave** and **Transpose** (Keyboard card, or Z / X keys) shift what you hear and record.
- Without MIDI: use the computer keyboard (A W S E D F T G Y H U J K O L P ;) — the letters are shown on the piano — or click the keys. Z / X change octave.

No sound? Browsers keep audio off until you interact with the page. Click anywhere (the orange banner says so) and play again.

5. Choosing and designing sounds

The **Instrument** panel has three tabs (Simple, Pro and Sampler — the Sampler is covered in its own section). Whatever you hear while playing is exactly what gets recorded.



Simple tab: pick an instrument-style sound.

- **Simple** — about 55 ready-made sounds in categories: Reference, Keys, Lead, Bass, Pad & Strings, Pluck & Mallet, EDM, Drums & Hits, SFX. Press a card to select it (a short preview plays). Cards marked “play low notes” (kick, sub drop ...) sound best around C1–C2.
- Three macro sliders adjust any sound: **Brightness**, **Length**, **Space** (reverb).
- **Save current sound** stores your own version under **My sounds** (kept in this browser, shared by all projects).
- **Reference** holds sounds built from your audio or descriptions, e.g. **Ghost Choir** — a soft, slowly rising choir with a gentle trembling (about 3.4 Hz vibrato and tremolo), wide stereo and a long reverb.
- The sounds are synthesized, so pianos and strings are approximations rather than sampled recordings.

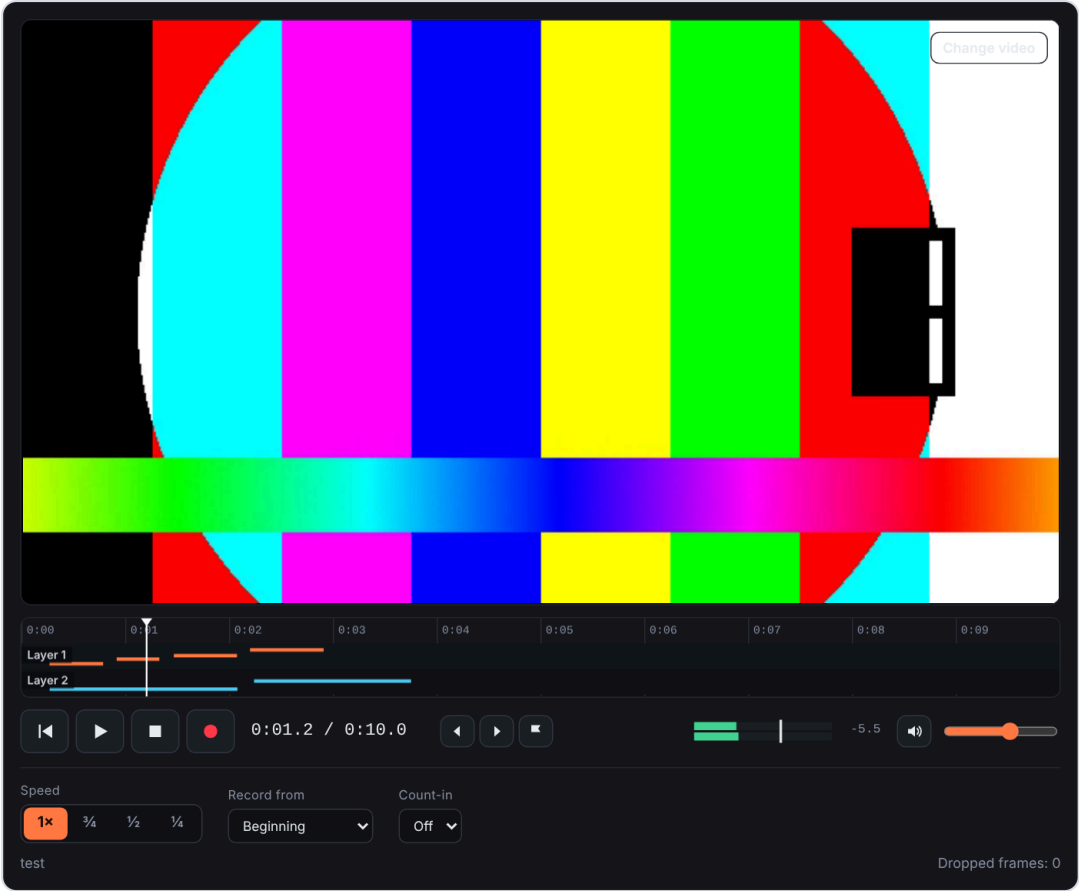


Pro tab: full synthesizer parameters.

Group	What it does
Oscillators	Two oscillators (sine / triangle / saw / square), the second with pitch and detune; plus a noise source.
FM	A modulator that makes bell, electric-piano and metallic tones (amount, ratio, decay).
Filter	Low-pass / high-pass / band-pass with cutoff, resonance, key tracking and a decaying envelope (negative amount sweeps up).
Amp envelope	Attack, decay, sustain, release of the volume.
Pitch & vibrato	Pitch envelope (zaps, drops), vibrato and tremolo (rate/depth), pitch-bend range.
Output	Stereo spread, reverb amount, level.

- The **Preset** menu and **Randomize** button are in the Pro tab. Editing any value switches the sound to “Custom”.
- On a layer, **Use this sound** copies the layer’s sound into the panel; **Apply instrument** replaces the layer’s sound with the panel’s sound — so you can improve a sound after recording.

6. Recording takes and layers

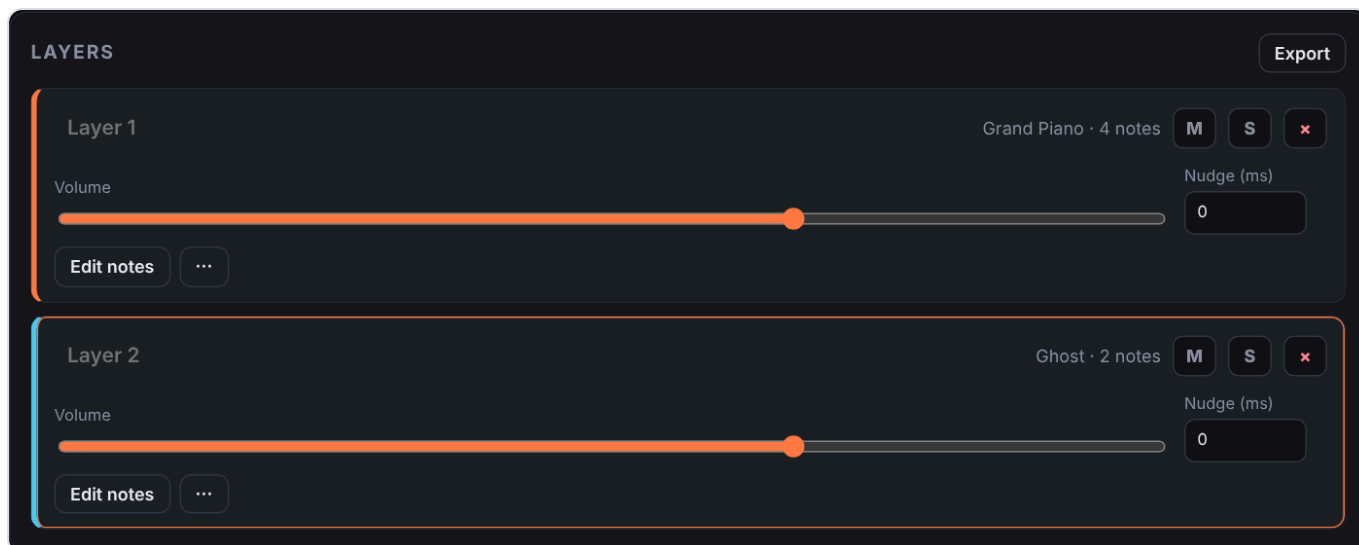


Player, timeline and transport.

Setting	Meaning
Speed	Playback speed while recording: 1 ×, ¾, ½ or ¼ (see the next section).
Record from	Beginning rewinds to 0:00 first. Current position starts where the playhead is.
Count-in	Seconds of count-down (with beeps) before the video starts. Off / 1 / 2 / 3 / 5 s.

Latency compensation and **Hear existing layers while recording** live in **Settings** (the slider icon at the top right).

1. Press **Record**. The video plays and the REC badge appears.
2. Play. Press **Record** again, **Stop** or **Esc** (or let the video end).
3. A highlighted **New take** card appears: **Preview** plays it with the video, **Keep as new layer** saves it, **Record again** discards it and records anew, **Discard** throws it away.



Layers: volume, mute / solo, nudge, export.

- Each layer has its own sound, **Volume**, **Mute**, **Solo**, **Nudge (ms)** (shift the whole layer in time), a name field, **WAV** (export only this layer) and **×** (delete).
- The timeline under the video shows all layers; click or drag it to scrub the video.
- While a take is waiting to be kept or discarded, the Record button is disabled — decide first.

7. Note editor (piano roll)

After recording, fix timing and length while watching the picture. Press **Edit notes** on a layer (a new take is selected automatically).



The note editor with one note selected.

Action	How
Move a note	Drag it left/right (time) and up/down (pitch). Notes snap to the chosen grid.
Change length	Drag the left or right edge of a note.
Add a note	Double-click an empty spot (default 250 ms).
Select several	Shift-click, or drag a box on empty space. Dragging moves all selected notes together.
Delete	Select and press Delete / Backspace, or the Delete button.
Nudge	Arrow keys: ← → 10 ms, ↑ ↓ one semitone; hold Shift for 100 ms / one octave.
Exact values	With one note selected, type Start (s), Length (ms), Pitch and Velocity below the roll.
Undo / Redo	Buttons, or Ctrl/Cmd + Z (Shift for redo).
Scrub the video	Click or drag the ruler at the top of the roll. Moving or selecting a note also jumps the video to that note's start, so you can judge sync by eye.
Zoom / scroll	Zoom slider or Ctrl/Cmd + wheel; wheel scrolls in time; "Fit" frames the layer; "Follow playhead" scrolls during playback.

Copy / paste / duplicate	Ctrl/Cmd + C copies the selected notes, Ctrl/Cmd + V pastes them at the playhead, Ctrl/Cmd + D duplicates them right after the selection, Ctrl/Cmd + A selects everything. The same actions are buttons under the roll.
Transpose selection	The -12 -1 +1 +12 buttons move the selected notes by octaves / semitones.
Velocity lane	The strip under the roll shows every note's velocity as a stem. Drag a stem up or down (all selected notes move together), or drag across empty space to paint velocities. Untick "Velocity lane" to hide it.

Edits made while the video is playing take effect when you release the mouse.

8. Slow-motion recording, frames and markers

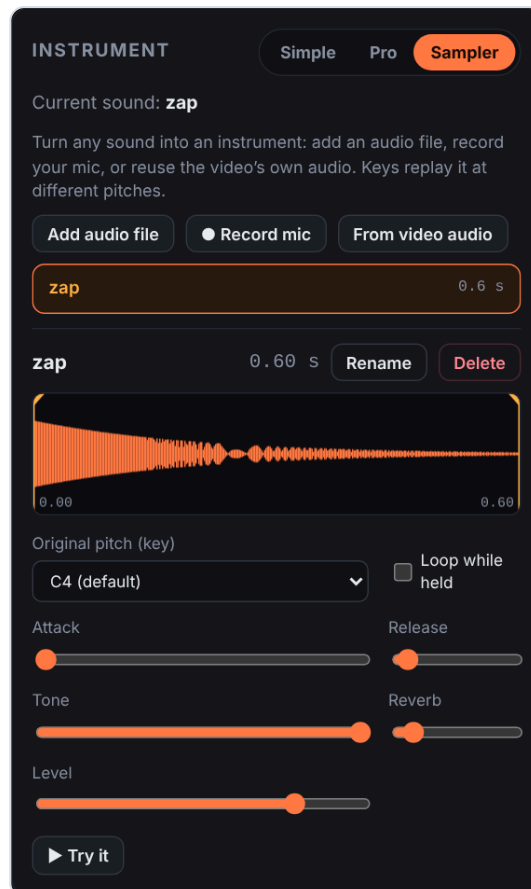
Tricky moments are easier to score slowly. Notes are stored in **video time**, so a take played at half speed lines up perfectly at normal speed.

Control	What it does
Speed 1 × $\frac{3}{4}$ $\frac{1}{2}$ $\frac{1}{4}$	Playback speed for playing and recording (under the player; locked while recording).
◀ ▶ buttons · , and . keys	Step one video frame back / forward (Shift = ten frames). The frame rate is detected automatically. With no note selected, ← → do the same.
Flag button · M	Drop a marker at the playhead; press again on the same spot to remove it. Markers appear on the timeline and in the note editor, notes snap to them, and they are exported to MIDI.
[and]	Jump to the previous / next marker.
Flag on the timeline	Click to jump there · double-click to rename · Shift-click to delete.

Latency compensation is applied in real time, so it stays correct at slow speeds.

9. Sampler: turn any sound into an instrument

The **Sampler** tab in the Instrument panel makes footsteps, a door slam or your own voice playable from the keyboard.



The Sampler tab with a loaded sound.

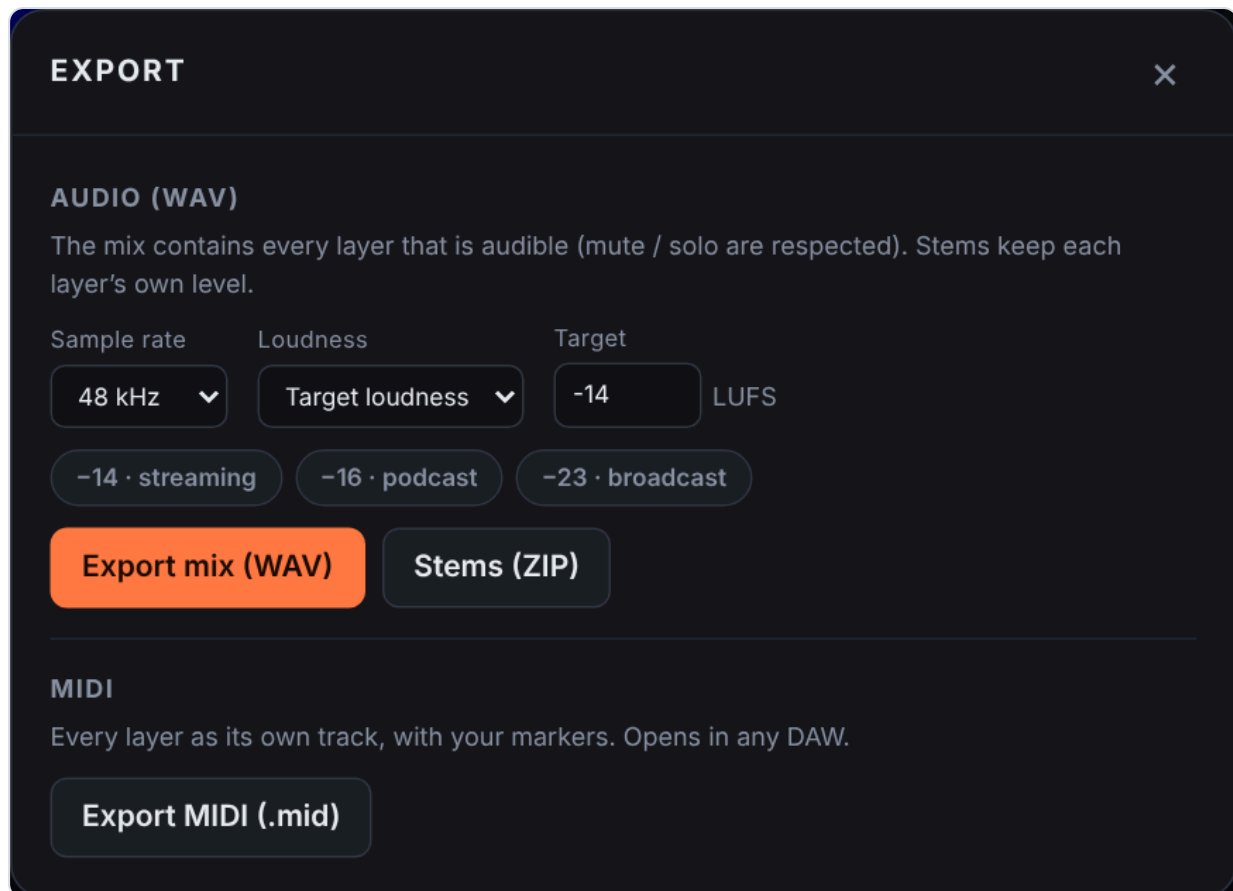
- **Add audio file** (WAV, MP3, M4A, OGG ...), **Record mic** (up to 30 s; the browser asks permission) or **From video audio** (the loaded video's soundtrack).
- Select a sample to see its waveform. Drag the orange handles to trim the start and end.
- **Original pitch** is the key that plays the sample unchanged; other keys play it higher or lower. Choose *No pitch change* for one-shot sound effects. **Loop while held** sustains the trimmed region.
- Attack, Release, Tone, Reverb and Level shape the result. A sampler sound works like any other: record with it and each layer remembers its sample.
- Samples are stored in your browser and shared by all projects. Backups (ZIP) include the samples a project uses.

10. Why video stutter does not throw things out of sync

Every note is stored against the **video's own clock** (its current time), not the computer's clock, and playback of your layers is scheduled against that same clock. If the browser stalls and a 10-second video takes 12 seconds to play, the picture and your music stall together, and recorded notes still land on the right frame. Export is rendered offline from those video-time positions, so the audio file is exactly as long and as aligned as the video. The "Dropped frames" counter under the player shows how rough playback was.

If a performance feels consistently early or late, use **Latency compensation** (for the whole take) or **Nudge** (for a layer), then fine-tune single notes in the editor.

11. Exporting audio and MIDI



The Export dialog.

- Press **Export** (top right). **Export mix (WAV)** renders all audible layers (Mute/Solo and volumes respected) as a 16-bit stereo WAV at 44.1 or 48 kHz.
- **Loudness:** *Off* keeps the level as played. *Peak -1 dBFS* raises the loudest peak to -1 dBFS. *Target loudness* measures integrated loudness (LUFS, ITU-R BS.1770) and adjusts it to your target — -14 for streaming, -16 podcasts, -23 broadcast — with a limiter keeping peaks under -1 dBFS. The measured result is shown after export.
- **Stems (ZIP)** exports every layer as its own WAV at its original level. A layer's ... menu exports just that layer as WAV or MIDI.
- **Export MIDI (.mid)** writes every layer as a track (timing matches the video 1:1) together with your markers, for any DAW.
- The level meter beside the volume control shows the master output; click it to reset the peak. It turns red when the signal reaches 0 dBFS.
- To combine with the picture, put the WAV and the original video on a timeline in any video editor, both starting at 0:00.

12. Backups, MIDI import, offline use and touch

- **Project → Export backup (ZIP)** bundles the project, its video and its samples into one file. **Restore from backup** (in that dialog, or *Import backup / file* on the project list) creates a new project from it — ideal for another computer or a safe copy. Every project card also has a **Backup** button.
- **Project → Import .mid** adds the tracks of a MIDI file as new layers using the current instrument (markers are imported too).
- **Install as an app:** use the install icon in Chrome / Edge's address bar, or “Add to Home Screen” on a tablet. After one online visit the app also works offline.
- **Touch screens:** the on-screen keyboard supports several fingers and sliding between keys; touching nearer the bottom of a key plays louder. On narrow screens it shows three octaves with < > buttons to move.
- The keyboard icon at the top right lists every shortcut (or press ?).

13. Keyboard shortcuts

Key	Action
Space	Play / pause
Esc	Stop (or stop recording)
Home	Go to start
, and .	Step one frame (Shift: ten)
M	Add / remove marker
[and]	Previous / next marker
Z / X	Octave down / up
A W S E D F T G Y H U J K O L P ;	Play notes with the computer keyboard
Ctrl/Cmd + C · V · D · A	Copy · paste · duplicate · select all notes
Delete	Delete selected notes
← → ↑ ↓	Nudge selected notes (← → step frames when nothing is selected)
Ctrl/Cmd + Z	Undo (Shift: redo)
?	Show the shortcut list

14. Troubleshooting

Problem	What to try
No sound	Click the page once (audio is blocked until then). Check the system volume and that the layer/instrument level is not zero.
“Web MIDI is not supported”	Use Chrome, Edge or Opera on a computer. Safari does not support Web MIDI.
Keyboard not listed	Re-plug it, close other apps that hold the device, press Connect MIDI again, and check the browser’s MIDI permission in the address-bar site settings.
Video will not play	Use MP4 (H.264) or WebM. Some MOV/MKV files use codecs the browser cannot decode — convert them first.
Notes feel late	Raise Latency compensation, or use Nudge on the layer.
Project missing	Projects are stored per browser and device. Use Export file / Import project file to move them.
Video gone after reload	The browser may have refused to store it (storage full). Load it again; free space or export a backup file.

15. Notes

- Everything runs locally in your browser; nothing is uploaded or sent to any server.
- Sounds are generated by a built-in synthesizer (Web Audio). Recorded audio is yours.
- Ctrl/Cmd+Z in the editor undoes note edits only; deleting a layer or project cannot be undone.